SELECT, WHERE and JOIN

SQL Server Essentials — Week 1 lecture notes

1. Reading data with SELECT

```
SELECT Title, Price
FROM Courses
ORDER BY Price DESC;
```

2. Filtering with WHERE

```
SELECT Title, Price
FROM Courses
WHERE Status = 'OpenForEnrollment' AND Price < 1500;
```

3. Joining tables

```
SELECT c.Title, cat.Name AS Category
FROM Courses c
JOIN Categories cat ON cat.Id = c.CategoryId;
```

Join	Returns
INNER JOIN	Only rows that match in both tables
LEFT JOIN	All rows from the left table, matches from the right (or NULL)

4. Grouping

```
SELECT cat.Name, COUNT(*) AS CourseCount
FROM Courses c
JOIN Categories cat ON cat.Id = c.CategoryId
GROUP BY cat.Name;
```

Practice

- List all courses with their instructor's name.
- Count enrolled learners per course.
- Find courses with no applications (hint: LEFT JOIN ... WHERE ... IS NULL).